Requirements Engineering Out of the Classroom: Anticipating Challenges Experienced in Practice

Pedro Marques*, Murilo Silva*, Camila Gusmão*, Diego Castro*† and Marcelo Schots*†
*Computer Science and Informatics Department, †MARVEL-SE Research Team
Rio de Janeiro State University (UERJ), Rio de Janeiro, RJ, Brazil
Email: pedroam27@hotmail.com, {murilojssilva, camilagusmao08, diegocbcastro07}@gmail.com, schots@ime.uerj.br

***Abstract*—Requirements Engineering (RE) education makes use of different methods to motivate and prepare students to perform well this activity in the practical scenario. Such educational methods, however, are typically based on some simulation, be it in the scope of the scenario, the problem, or the stakeholders. A change in the traditional teaching paradigm becomes necessary, bringing the student to the center of the RE activities. This helps them realize the importance of properly identifying and organizing the demands from the target audience, so that a quality requirements specification document can be produced. This paper describes a method applied during the Requirements Engineering course using a real scenario, in which students sought to identify the diverse stakeholders' needs with respect to the university institute website. Through this work, students achieved effective engagement in practical RE activities, whose results has benefits not only to the university community, but also to the external public interested in institute information. The conduction of this work revealed – and reinforced – some relevant challenges and lessons learned.**

***Index Terms*—Situated learning, requirements engineering, software engineering education**

I. Introduction

A recurring challenge in many areas of education (most notably in Computing) is reconciling theory and practice, providing the required knowledge during the learning process for enabling students in succeeding in their chosen field of occupation. In Software Engineering, and more specifically in Requirements Engineering (RE), making the academic experience more appropriate to the practical application in the field has been an issue for a while [1]. Some reports on RE education try to better approach the practical side with activities based on simulated scenarios (*simulated real world experience*) [2] or through role playing [3].

However, such a simulation does not provide a complete immersion in the RE scenario, due to its underlying limitations [4]. Some elements that constitute the real scenarios are difficult to explore or reproduce, such as uncertainties of requirements suppliers as to what should be developed, conflicting requirements, behavioral traits, among other technical and non-technical aspects [5].

Considering the importance of the university as the provider of human resources capable of dealing with the most diverse challenges in the real world, this paper presents a practical approach which aims at bringing the student to the center of the RE activities using actual university demands as a background. The context is focused on a particular setting: identifying and understanding the stakeholders' demands to be met in the adaptation of a university institute website.

This work is strongly based on the principles of Situated Learning [6], an educational theory based on the Learning by Doing approach. According to Navarro et al. [7], while Learning by Doing centers its focus on specific learning activities that students must accomplish, Situated Learning cares more about the environment in which learning occurs, based on the idea that knowledge is situated, being mostly a product of the activity, context, and culture in which it is used. Therefore, the environment in which students practice their recently acquired knowledge must be "authentic", i.e. an environment as similar as possible to the one in which the knowledge will be used [7].

The remainder of this paper is organized as follows: Section II brings the context and motivation for this work; Section III describes the applied method; Section IV lists some challenges faced and lessons learned during its conduction; Section V discusses related works; and Section VI lists some final remarks.

II. Context and Motivation

The Institute of Mathematics and Statistics (IME) begun its activities in 1968, and nowadays has hundreds of professors and administrative technicians, besides collaborators and other staff members. In the undergraduate level, it currently offers one Licentiate Degree and four Bachelor Degrees, whereas in the graduate level it counts with four graduate programs (*lato sensu* and *stricto sensu*). There is also a variety of information about news, events, extension projects, research opportunities, and so on.

Aiming at making such information accessible and reachable, also advertising the IME activities, the institute has a website[1], whose most recent version was created six years ago, based on an existing one. Even though the undergone restructuring was naturally instigated by several limitations of the discontinued version, one could not assure that the structure and information currently available meet the expectations of the target audience. Furthermore, professors and students frequently point out some improvements, to be analyzed by the webmaster. A common (acceptable) reason for not following some suggestions is the fact that it should be validated with other stakeholders; this would then justify the endeavor.

[1]UERJ's Institute of Mathematics and Statistics: http://www.ime.uerj.br/

978-1-7281-6807-4/20/$31.00 ©2020 IEEE

Authorized licensed use limited to: IEEE Xplore. Downloaded on November 01,2020 at 18:08:15 UTC from IEEE Xplore. Restrictions apply.

As in any software development project, the identification of the information needs and features for the institutional website must be performed in the scope of a requirements analysis and specification activity. This is explained in the Software Engineering course, and taught in greater level of details in the Requirements Engineering course, both being part of the Computer Science Course Degree.

One of the toughest problems pointed out in the formation of Computer Science professionals is the possible lack of preparation towards the practical side of the occupation area [1]. Thus, it would be interesting to integrate a *demand for solutions* (from the target audience of the institutional website), the *learning demand* (from the students) and the *demand for qualified professionals* (from the industry) in a project that puts students in the center of RE activities, under the supervision of a professor.

An important benefit of this setting is the possibility of a more flexible adaptation to the reality of the students, provided by the university environment, bringing interesting learning challenges without the existing pressures in the industry.

Even though the project made use of an academic domain, somehow familiar to the students, it represents an opportunity to explore many challenges of RE, since:

(i) students (especially undergraduate ones) are not familiar with all the intricacies of the academic scenario, so that there are novel aspects to be explored;
(ii) in the RE education scope, working upon domains completely unknown to students can be too challenging considering the teaching and learning objectives;
(iii) lastly, the challenges of dealing with conflicting requirements, indecisive stakeholders etc. are not restrained to some domains, occurring in a wide range of scenarios.

Additionally, the topics and scope of the project are defined solely by the professor, without any involvement of the students, similarly to the industry scenario, in which one has no control over which project they are part of, and does not work based on personal preferences.

The evaluation of students during the course is not focused on written exams, but based on (i) the amount of completed tasks, (ii) the time it took for them to do so, and (iii) how well tasks were completed. Non-technical aspects (e.g., level of self-confidence to conduct tasks involving stakeholders) are not subject to grade, but worked individually instead, so that students could perform better in the next stages.

III. Method Used

This work followed the stages shown in Fig. 1, detailed in the next subsections.

A. Activity Planning

During the activity planning, it was decided that, whenever possible, the practical tasks should take place immediately after the theoretical classes, taking into account the academic schedules for not harming other student activities (e.g., other courses, monitoring, and technological/scientific initiation scholarships).

The individuals involved in the website initiative were instructed to prepare the students for the practical scenario without giving up on the necessary realism, i.e., without acting towards the students' education as professors usually do. They understood the proposal and its goal, promptly agreeing to take part in the activity.

B. Introduction to the Course Content

Five undergraduate students took part in the project. They all had some knowledge of RE from the Software Engineering course, and little or no practical experience in RE, but proving themselves capable and interested in completing the several tasks required in the project.

The classes had a highly expository nature, guided by slide presentations and long conversations with the students They were instigated to solve problems and carry out assignments related to the target domain, ensuring that they would have a good assimilation of the course syllabus.

The course content started with a brief introduction to RE, followed by a recap of requirements classification (e.g., functional, non-functional, and business requirements), instructions on how to conduct RE correctly, and an in-depth explanation of various requirement acquisition techniques, some of which were used in this project. Some cases were also presented based on the professor's previous experiences in the practice of RE processes in software development organizations.

With a weekly workload of roughly 3 hours during a semester, the classes were held once a week, introducing and explaining the theory that would serve as a basis for the project, with new assignments being issued weekly to be completed before the next class. Most classes also focused on aspects related to the practical application of RE.

C. Domain Analysis

Before effectively communicating with the stakeholders, students needed to get a better understanding of the project domain, which would be useful for further stages in the requirements acquisition and specification. Since the goal was to create a new version of an existing website, the understanding of the domain was achieved through two steps:

(i) the mapping of the website domain from other academic institutions, looking for patterns and common functionalities that could be requested (i.e., features relevant to the domain); and
(ii) an analysis of the structure of the current version, seeking to understand the existing features (comparing with the performed mapping) and the detection of problems, omissions, and areas of improvement.

The analysis sought to identify how some technological institutes used their website to convey and provide information to their target audience. Some websites stood out by providing important features with intuitive interaction.

After finishing these steps, the students created feature models [8] using the FODA (Feature-Oriented Domain Analysis) notation [9], with the goal of organizing the identified information. Fig. 2 presents an example. These models were

Authorized licensed use limited to: IEEE Xplore. Downloaded on November 01,2020 at 18:08:15 UTC from IEEE Xplore. Restrictions apply.

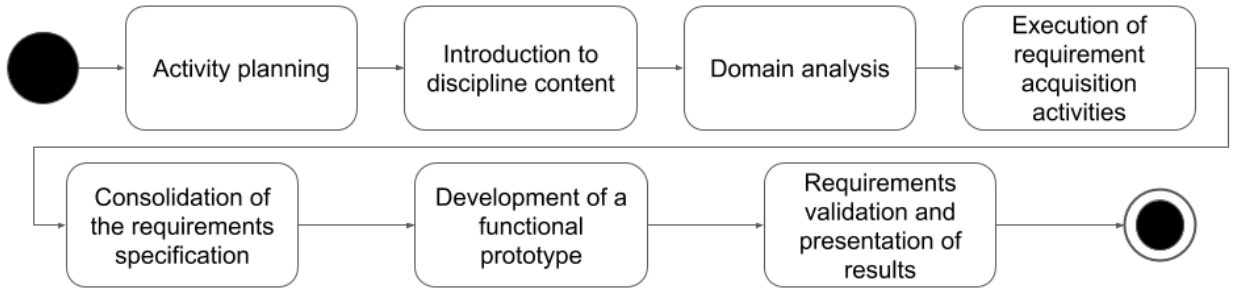

Figure 1. Method used in the context of this work

then combined and refined based on some outputs from the further stages, as well as the professor's and some stakeholders' feedback. This served as a basis for the organization of following stages. Some model elements include:

- **About Us**: A brief introduction to the history of the institute, from its creation to the current structuring;
- **Home page**: An area showing the most recent institute news, an animated banner ("carousel") showing highlighted news, and an area for the next institute or university events;
- **Teaching Staff**: Information about the professors linked to the institute, such as their research areas and their contact information;
- **Undergraduate Course Degrees**: Information about the course degrees offered by the institute, e.g., their curriculum, courses, and the coordinator;
- **Contact Us**: Contact information, e.g., phone, work address, and e-mail;
- **Responsive Layout**: A layout that adapts the website visual elements to different devices, e.g., mobile (a domain in which RE has been neglected [10]).

D. Execution of the Requirement Acquisition Activities

Requirements elicitation is a challenging task to which students must be properly prepared [11]. To that end, students were trained in requirements acquisition techniques through presentations, rehearsals, and Q&A challenges. The students had to develop interview questions, diagrams, and scripts to make sure that the meetings with the stakeholders (and the other project activities) could be carried out as smoothly as possible. This allowed to observe difficulties commonly experienced by students when performing elicitation activities for the first time, as well as some ways of approaching these to make the process more comfortable and dynamic to all involved. The following subsections present the techniques used in the project.

1) Initial Interview: In order to start the first step of the requirements acquisition stage, an interview was scheduled with a key project stakeholder (one of the main interested persons on the initiative to develop a new website version). The elaboration of questions to this stakeholder was one of the crucial points in this technique, because of the need to detect what should be asked (and what should not) in the available time, and what would be the best way to gather appropriate answers, while also checking the topics that the stakeholder could effectively discuss.

Since this was the first contact with a stakeholder, the questions were organized according to their degree of generality and grouped by topics (including generic topics and others derived from the feature model). Afterwards, it was necessary to define the sequence of questions for properly connecting the topics, easing the meeting flow. Examples of questions include: "What drawbacks do you see in the current version of the website?", "Is there something in the website that should not be?", "Who is currently in charge of the website maintenance?", and "What sort of content external to the institute the website should contain?". Establishing an ideal sequence, however, was not an easy task.

During the classes, students analyzed some requirements acquisition techniques and defined the questions would be posed during each part of the interview. They started asking open-ended questions to stimulate more descriptive answers and encourage discussion, and begun performing the interview based on the funneling technique[2]. To obtain relevant information more easily, a key aspect taken into account was to keep the stakeholder as comfortable as possible.

The professor (serving as a facilitator) supervised the interview conducted by the students, who took turns between asking questions and note taking. The interviewee showed much engagement in the project initiative, becoming a useful source of information about the envisioned website functions. At certain times, the interviewee ended up anticipating answers to some still unasked questions while answering a previous one, causing the need to change the previously planned interview structure on-the-fly, requiring perception and communication among the students.

At some point of the interview, the interviewee stated that he prepared a sketch beforehand, based on how he envisioned the website. Due to this unexpected situation, the interview format had to change on-the-fly. Although some of the required content envisioned by the stakeholder were outside the scope

[2]Funnelling is an interview technique in which most general questions are asked first, and more specific questions are asked as the interview goes on [12].

Authorized licensed use limited to: IEEE Xplore. Downloaded on November 01,2020 at 18:08:15 UTC from IEEE Xplore. Restrictions apply.

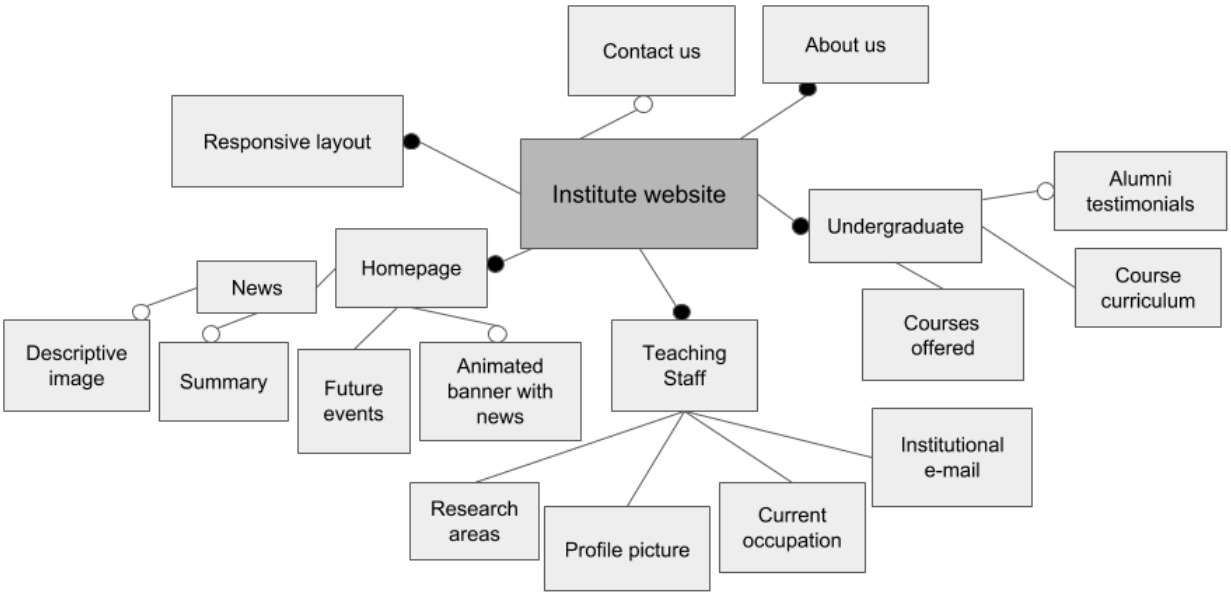

Figure 2. Developed feature model (using a radial layout)

of the project (characterizing a new system), it was decided not to start a requirements negotiation at that moment. Through this interview, it was possible to generate a first version of the requirements document.

2) Brainstorming: The first interview provided a clearer picture of the problems and necessities to be developed for the institutional website. By analyzing the information collected during the interview, it was possible to figure out aspects that were missing or needed further clarification, as well as new inquiries that emerged. Thus, to generate a new version of the requirements document, a new meeting was scheduled in the format of a brainstorming [13], seeking to obtain complementary information from more stakeholders while they freely debated about each topic. A positive aspect of this approach is that the participants argued about aspects that they considered important for the website, while also exposing new ideas that were not previously explored or thought of.

This technique also required appropriate preparation. The students created a concept map to serve as a guide during the interview, containing relevant topics to the brainstorming, as well as some specific topics that should be debated upon. This allowed to obtain a more comprehensive view of the aspects to be addressed in the website, as well as their interconnections, also helping in the order in which the topics should be raised to facilitate the conduction of the brainstorming.

The meeting worked as a large group conversation, in which participants could interact with each other on the same topics. Because of the number of participants (four stakeholders), it was decided that the professor would take the role of moderator, in order to establish the topics for discussion and intervene if/when necessary, to prevent the meeting from going off-topic. Topics ranged between pros and cons in the current website version, information to be presented on the home page, kinds of news to be published in the website, among others. It was possible to obtain an overview of other perceptions and opinions about the project, as well as punctual discrepancies between the stakeholders' viewpoints (some of which were perceptible from the beginning), ending up reaching a consensus by the end of the brainstorming. After the activity was concluded, some early stage requirements negotiations were established, using soft negotiation strategies [14].

Because the project was still in a stage in which the scope could be altered more easily, the brainstorming proved to be very productive. Besides, it became possible to pre-validate some requirements, as well as detect new ones, while also getting answers to previous inquiries. This activity proved to be very useful and important for the generation of the second version of the requirements document.

3) Questionnaire: The previous requirements acquisition techniques targeted a few professors. For a broader identification of other points of view, it is necessary to query another important group of stakeholders: the students. Since their amount is massively bigger than that of the teaching staff (in the thousands), performing interviews or brainstorming sessions would be costly and inefficient. To overcome this, the students elaborated a questionnaire, aiming at achieving a bigger reach and making this process more accessible for the students willing to participate.

A form was created with multiple choice questions and grid scales for some answers, adjusting itself dynamically according to the answers of each respondent. For instance, if the respondent has never visited the current version of the website, no questions related to this aspect are shown. Although the multiple choice format limits the scope of answers, the idea was to speed up the comprehension and analysis of the collected data. Yet, due to some setbacks on

Authorized licensed use limited to: IEEE Xplore. Downloaded on November 01,2020 at 18:08:15 UTC from IEEE Xplore. Restrictions apply.

the university schedule and because of time frame restrictions, it was necessary to limit the scope of participation. A new round is planned to gather feedback in the future, considering the evolved version of the website.

4) Prototyping: To ensure that the understanding of the requirements obtained in the previous stages was correct, the students elaborated a prototype, based on the most recent version (by then) of the requirements document, representing the system concepts and functionality. To demonstrate the design of the pages and the information flow, students elaborate some sketches, inspired by the layout envisioned by one of the stakeholders after the interview stage.

With the prototype, it was possible to obtain and suggest aspects to be improved earlier in the project, therefore preventing the risk of detecting them in further stages. It helped solving some inquiries and clarifying some issues, providing a clearer picture with respect to the placement of information in the website, contrasting with the previously obtained information. The prototype was also useful for discussing how some information would be visually presented.

The purpose of this prototype was not to verify quality or security aspects of the website, but to validate with the stakeholders whether the expectations were being met and what required further analysis to reach a satisfactory result.

E. Consolidation of the Requirements Specification

The requirements specification document was elaborated with the goal of organizing all the information obtained from the previous steps. It contains, among others: (i) the *project scope*, detailing which aspects would be part of the project, and which would not; (ii) the *description of the requirements acquisition techniques*, detailing each of the techniques employed for requirements acquisition (as mentioned previously); (iii) the *dictionary of terms and abbreviations* used in the document, along with their meaning and references; (iv) the *system actors*, detailing the user types that will hold any role in the system; (v) the *functional requirements*, *non-functional requirements* and *business rules*, grouped by related topics; and (vi) the *references*, listing the papers and resources used in the specification.

To better define the role of each system actor, as well as their different access levels over the website functions, a use case diagram was elaborated, as presented in Fig. 3. The *Visiting User* can view any public content on the website, while the *Authenticated User* has the permission to manage their own profile and access the content restricted to his/her role. This type of actor, who is normally a professor, can also be a *Content Administrator*, who can manage the news published. All these roles are registered by *System Administrator*, who has permissions to manage the website as a whole.

F. Development of a Functional Prototype

Some versions of functional prototypes were developed based on the requirements specification document. The beginning of the implementation was made during a course (Development with Ruby on Rails) offered in the following semester, taught by another professor from the same department (who was also engaged in having a new website). During the course time frame, two other students not enrolled in the RE course joined the project development. At this point, the existence of a previous requirements specification proven to be particularly useful. The prototype kept evolving based on the views of the project stakeholders.

Firstly, the home page, news, departments, and professors views were implemented, followed by the header (which contains the menu that redirects to all the other views), and the footer (which contains the contact information and hyperlinks to websites related to the institute). At this moment, the aesthetic of the website was not targeted, but its functionality instead. The inclusion of a carousel resource complemented the Home page, responsible for the dynamic display of highlighted news about the institute and its departments, along with a listing of the last news published. The Content Administrator role is responsible for choosing the content to be highlighted and the period during which it remains so.

The Professors, Staff, Departments, and Course Degree pages display a listing with all their corresponding entities, each including a description or profile. The System Administrator must register each of these entities, while the Authenticated User can manage his/her own profile (displayed on the professors or staff view). The prototype allowed to define more clearly how this information would be organized in the layout. Fig. 4 and Fig. 5 present, respectively, the Home page and the Department page.

The evolution of the prototype was hosted on a GitLab version control and issue tracking systems, which allowed to organize what parts of the website each student would be responsible for implementing. The idea was to develop an evolving prototype, which could be gradually implemented based on the evolution of requirements (in future meetings with the stakeholders). In order to optimize the tests of each version of the prototype, the system was hosted at a virtual machine with different branches for each student. Thus, after making some modification and checking that it did not affect the system functioning, each student could merge it into the main repository with less error-proneness.

G. Requirements Validation and Presentation of Results

Before the beginning of this stage, the students presented the prototype to the professor of the Development with Ruby on Rails course, counting as the final grade of the class. Although the primary focus of this specific evaluation was on technical aspects, the intent to keep it as an evolving prototype was consolidated.

After that, the requirements validation was performed through a preliminary meeting with one of the stakeholders, followed by a presentation involving the board of directors. The stakeholders who attended these meeting analyzed the prototype and pointed out some areas of improvement. They were satisfied with the results, and also suggested the integration with other university systems, which may be done in future projects.

Authorized licensed use limited to: IEEE Xplore. Downloaded on November 01,2020 at 18:08:15 UTC from IEEE Xplore. Restrictions apply.

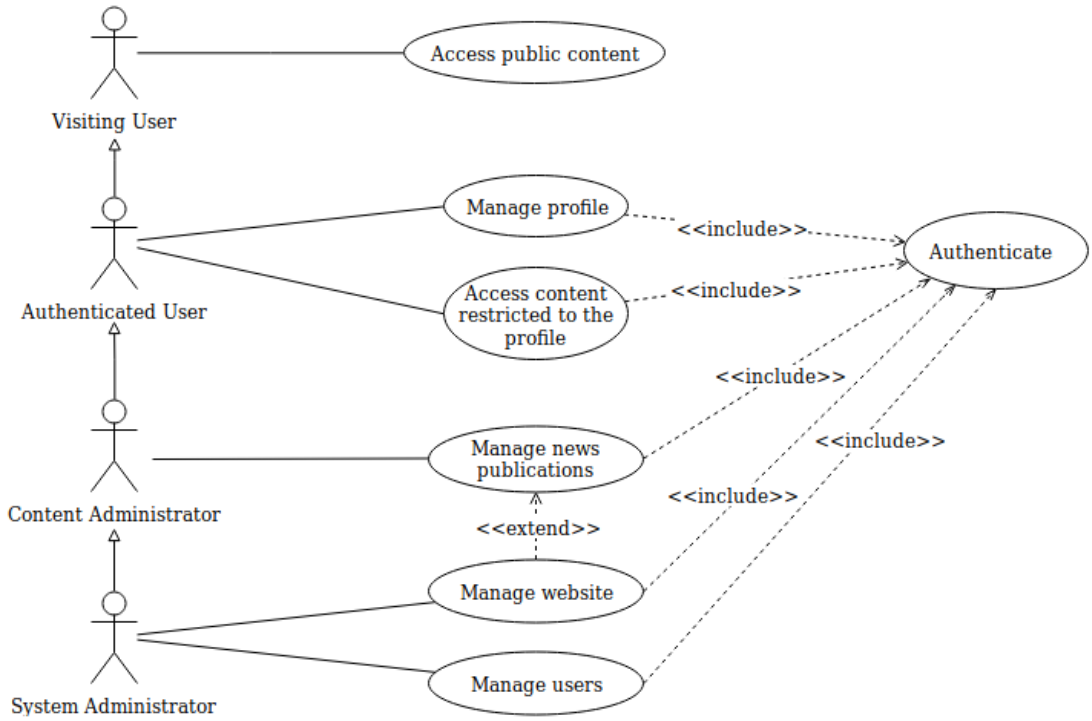

Figure 3. Use case diagram of the website project

IV. Identified Challenges and Lessons Learned

As in any software project, some challenges were faced during the conduction of this work, many of which meet the industry reality. There were also challenges related to the education setting, which must be handled in future initiatives. Such challenges and the learned lessons with them are listed and discussed as follows.

Acquisition and understanding of the requirements during the meetings: Students had to organize themselves (before and during the meetings) to gather as much information as possible, and to write them down in a short period of time. This is due to a foreseen limitation with respect to the available schedule of the stakeholders for the meetings and the brainstorming session. Since this was their first time dealing with such experience, these foreseen difficulties were managed over time. Each new meeting revealed a substantial evolution of the students' abilities, strategies, and critical thinking.

Conciliation of ideas and requirements management: The conflicting requirements identified in each new stage made evident the practical difficulty in conciliating ideas given by the stakeholders with the requirements obtained in the previous stages. This was a great opportunity to apply the inspection activities learned in the Software Engineering course. It was also easy to notice the difficulty for client stakeholders to recognize the project scope (as frequently noticed in the industry), bringing the need to negotiate requirements and establish possibilities of future projects. In fact, some ideas constitute projects of a different nature (e.g., an integrated virtual learning environment).

Comprehension of the system necessities: Because the stakeholders themselves did not have a clear understanding about what would be an ideal website at a user level, the project required a substantial amount of time in defining and modeling the organization and division of actors and their permission levels to fulfill the actual institute needs. The answers to these inquiries became clearer to all stakeholders after the elaboration of the prototype, when it was possible to illustrate the obtained understanding and suggest a possible mode of operation that would meet the institute needs. This operation mode ended up differing from the original proposal, but proved to be more effective.

Conciliation with other courses: The purpose of the Development with Ruby on Rails course was much more geared towards technical aspects of the project. Thus, the involved students had to prioritize certain actions to ensure a good grade. At that moment, priority was given to the implementation of specific prototype functions, which were not necessarily the most important ones for the validation stages. Furthermore, despite the facilities brought by Ruby on Rails and the availability of libraries (arguments considered by the students in its choice), the students' lack of familiarity with the framework largely affected the progress of the project.

Identification of the abilities and preferences of the students: Each student had different levels of knowledge, ability, and interest over different aspects of the project, such as organization of ideas, interpersonal communication, critical thinking, software implementation, among others. Some students were more introspective, while some were more communicative. These

Authorized licensed use limited to: IEEE Xplore. Downloaded on November 01,2020 at 18:08:15 UTC from IEEE Xplore. Restrictions apply.

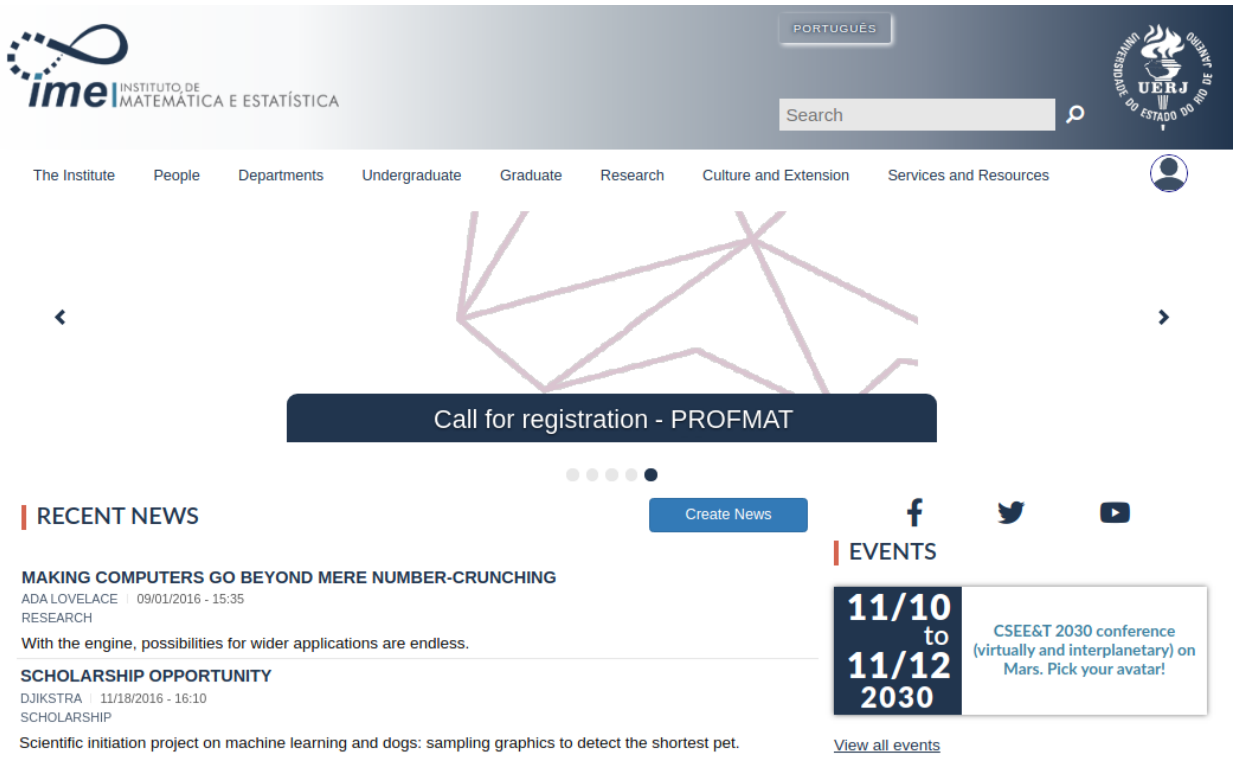

Figure 4. Prototype screenshot – Home page (with fictitious data, for illustration purposes)

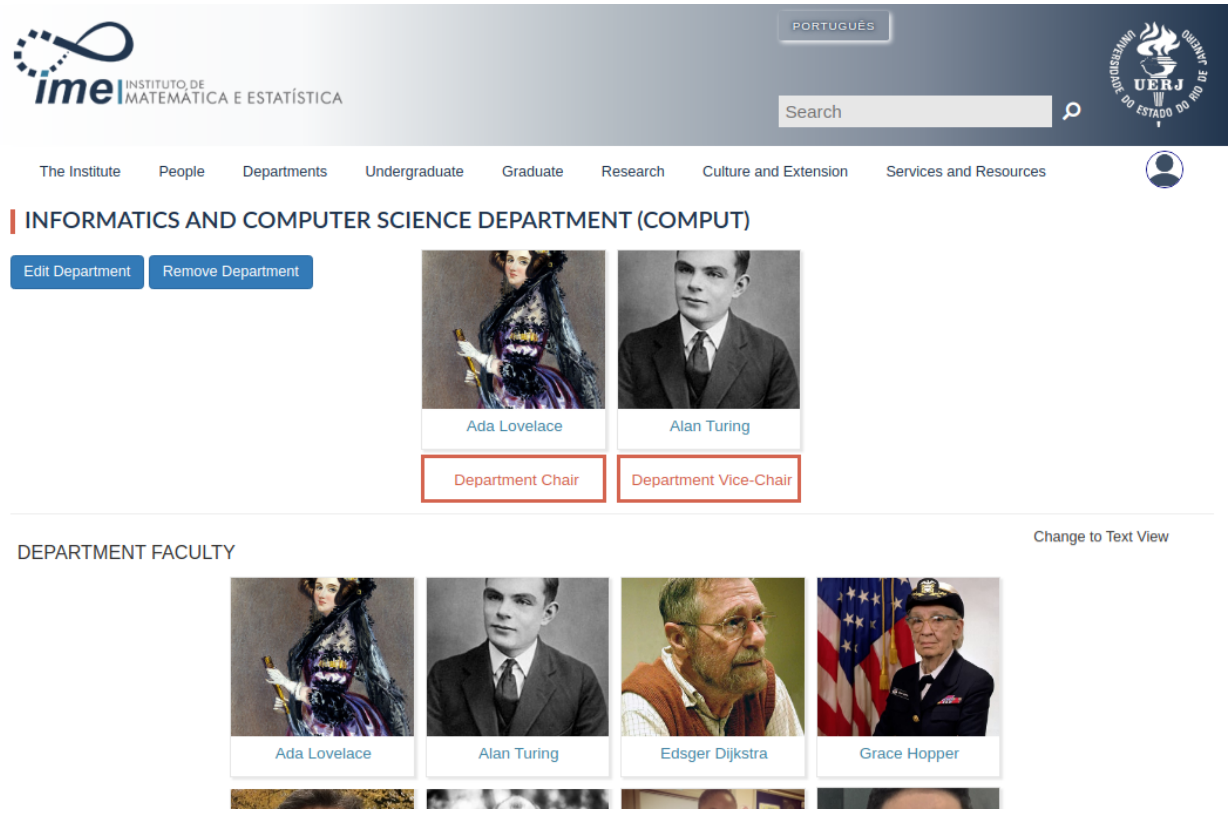

Figure 5. Prototype screenshot – Department page (with notorious Computer Science contributors not associated with IME, depicted for illustration purposes)

Authorized licensed use limited to: IEEE Xplore. Downloaded on November 01,2020 at 18:08:15 UTC from IEEE Xplore. Restrictions apply.

aspects were observed during the conduction of the project, and a strategy was adopted for making students to "get away from their comfort zone" (challenging them in developing new abilities) while "getting the best out of themselves" (benefiting from their potential in existing abilities). This led to some relocation of tasks, which also helped improving their progress in the project. The fact that non-technical aspects would not influence their grade also made them feel more comfortable to experience roles they would not easily agree with, allowing them to achieve great results. Such a flexibility is one of the greatest distinctions with respect to industry training.

Time and availability: The course duration, including the learning of the theory and the practical activities, proved to be too short to encompass all the intended stages, as mentioned previously. This was especially true when taking into consideration the students' availability during the project: one of the students began an internship, while the other (already in his last semester) got a job. This made it harder both the scheduling of meetings and the fulfilling of tasks within the deadline. Yet, in spite of the individual schedules, the students, the professor, and the other stakeholders involved in this initiative agreed that it was well worth the effort.

V. Related Works

Zowghi and Paryani [3] noted that engaging the students in hands-on activities would be beneficial to the learning and practice of the RE course. The students were divided into two groups: developer group and customer group. While some students assumed the role of consumers to a developer group, they simultaneously served as the developers to another customer group. The authors concluded that this practice was very useful for the students, because they gained a better understanding of the RE process and activities, along with a hands-on experience with modeling, analysis, validation, and management.

Macaulay and Mylopoulos [15] sought to analyze how RE was being taught at the universities (both at the undergraduate and graduate levels), contrasting with the abilities that industry required from future professionals of this field. To this end, they interviewed professors of some universities and sent surveys to some companies. From the results, the authors defined a list of topics related to RE that should be taught at the university level to make the students more aligned to industry expectations. This list was divided into two parts: a more theoretical one, focusing on RE concepts and techniques, and another involving tools and skills that are useful or necessary for practical requirements acquisition.

The work by Daun et al. [16] also pointed out the disparity between what was taught in classes and what the industry actually needed. The study proposed a different teaching style from the one used at the time, in which students learned the theory and then were trained in some software tools used in industry, while also making weekly group assignments proposing for a solution to a given problem. The proposed method sought to stimulate more interaction and discussions among the students, making them better prepared to work in larger groups. As a result, the authors noticed an increase in the interaction between the students, who ended up turning in more assignments, even the optional ones.

Memon et al. [4] applied a survey to students of Malaysian universities, requesting opinions about RE teaching. 40% of the respondents were undergraduate and the others were graduate students. More than half of the students stated that they believe they had learned the key points of RE during the course; however, only 50% considered themselves capable of using their knowledge in the industry. When asked about the biggest difficulties they had during the course, a frequent mention was the large amount of time focused on the teaching of theory, therefore causing a deficiency in the practical aspects, involving tools used in the industry and performing group projects.

Another relevant aspect pointed out by Mohan and Chenoweth [5] about the RE teaching curriculum was the importance of inserting the students in projects jointly with industry companies, based in scenarios that could evidence better where and how the techniques learned in the first part of the course should be applied. This experience resulted in 33 out of 36 projects being successful, and this success was accredited to the demonstration of how the theory should be applied and why it works, instead of using "toy" projects (i.e., without a real client, only for practicing the theory in class).

Our work differs from the others due to the following characteristics:

(i) The teaching was carried out giving emphasis to each stage of the project execution, from the theory to the usage of the requirements acquisition techniques and the actual development of the website;
(ii) Rather than choosing a "toy project" or test project, a real project was used, involving a significant demand for the students, which permitted a greater interaction of the student team with the stakeholders during the entirety of its execution, making it possible to obtain not only a more effective learning by means of the practical application of the concepts, but also a great level of commitment and engagement by the students;
(iii) Ruby on Rails was chosen for the development of the website, being a popular framework used by cutting-edge companies; its choice brought a direct contact with the problems and difficulties that can be encountered in a professional environment;
(iv) There was an extensive preparation to determine how the interviews and brainstorming would be carried out, making it possible to obtain not only an experience in the elaboration of questions and the flow of topics that would be discussed, but also aspects of hospitality, etiquette, and personal presentation that should be followed in these types of situations;
(v) The theory taught during the course (sometimes including parts of other courses) was applied in the website project, involving the creation of the feature model and the use case diagram, besides the division of the contents on each view and the website users' interaction flow.

Authorized licensed use limited to: IEEE Xplore. Downloaded on November 01,2020 at 18:08:15 UTC from IEEE Xplore. Restrictions apply.

VI. Final Remarks

RE is a fundamental stage in the development of quality software projects, allowing the construction of more trustworthy systems when done the right way, conveying the functionality, constraints, and rules that must be present and/or be accomplished. To prevent the error propagation to further stages in the development, the importance of executing RE correctly is getting increasingly noticed by the organizations, which often do not possess qualified workers in this field.

In many university courses, the classes are restricted to the display of concepts, without taking proper care to show their application to students, who end up replicating definitions without understanding properly their actual goal, meaning, and content. This creates professionals who, in theory, know how to apply the concepts, but in practice execute requirements acquisition techniques inadequately, generating shallow and incomplete documentation.

Students lack in the university some experience with scenarios that effectively put in practice the obtained knowledge, deemed to be forgotten otherwise. When the university courses allow students to put in practice what they were taught, the concepts become more solid and meaningful to them. This work exercised the fundamentals of RE with the students, who could realize its utmost importance and applicability.

A. Limitations

The main limitations (some already mentioned in Section IV) include the challenges in scaling this work to a large number of students, the lack of a tool specifically geared towards requirements management (not used in this work) and the lack of a key technical figure to help solving inquiries about the programming language and framework.

The students considered the adaptation time as a hurdle that needed to be overcome by those who never had any industry experience, since this project had a relatively complex nature, and was not limited to a simple didactic example for the consolidation of a theoretic knowledge. Apart from that, students ended up feeling more able to handle unforeseen events and manage risks. Their experience in previous undergraduate courses also played a role in this project.

B. Future Works

This work can provide some support to the application of this method in other educational settings, as a way of preparing the students to the job market. This practice is common in other courses of the institute, such as Software Engineering and Systems Analysis & Design. There are plans for assessing other teaching methods combined with different settings, in collaboration with other professors.

The conclusion of this project allows its further extension, in the creation and restructuring of other university websites, which contributes for a better communication and provisioning of information. Lastly, in future opportunities, one can handle different aspects of RE in practice, including the requirements traceability, as well as the integration of RE with other Software Engineering topics.

VII. Acknowledgments

The authors would like to thank the Extension Department of the Rio de Janeiro State University (DEPEXT/UERJ) for the opportunity in conducting this project. Prof. Marcelo Schots also thanks all the participant students, for voluntarily joining this initiative and collaborating to its fulfilment.

The authors would also like to thank the collaboration of the IME/UERJ students (especially Suelen Santos, Vinícius Soares, Daniel Coutinho, Lucas Tavares, Igor Silva, Paulo Victor Coelho, and Renato Domingues), professors (especially Prof. Americo Cunha, Geraldo Magela, and Leandro Marzulo), and the IME/UERJ staff as a whole. Finally, the authors thank Prof. Natália Schots (IM/UFRRJ) for her suggestions for the camera-ready version.

References

[1] J. A. Bubenko, "Challenges in requirements engineering," in *2nd IEEE International Symposium on Requirements Engineering*, ser. RE, March 1995, pp. 160–162.

[2] G. Regev, D. C. Gause, and A. Wegmann, "Requirements engineering education in the 21st century, an experiential learning approach," in *16th IEEE International Requirements Engineering Conference*, ser. RE, September 2008, pp. 85–94.

[3] D. Zowghi and S. Paryani, "Teaching requirements engineering through role playing: Lessons learnt," in *11th IEEE International Requirements Engineering Conference*, ser. RE, September 2003, pp. 233–241.

[4] R. N. Memon, R. Ahmad, and S. S. Salim, "Problems in requirements engineering education: a survey," in *8th International Conference on Frontiers of Information Technology*, 2010, p. 5.

[5] S. Mohan and S. Chenoweth, "Teaching requirements engineering to undergraduate students," in *42nd ACM Technical Symposium on Computer Science Education*, 2011, pp. 141–146.

[6] J. Lave, *Cognition in practice: Mind, mathematics and culture in everyday life*. Cambridge University Press, 1988.

[7] E. O. Navarro, "On the role of learning theories in furthering software engineering education," in *Software Engineering: Effective Teaching and Learning Approaches and Practices*. IGI Global, 2009, pp. 38–59.

[8] K. Lee, K. C. Kang, and J. Lee, "Concepts and guidelines of feature modeling for product line software engineering," in *7th International Conference on Software Reuse: Methods, Techniques, and Tools*, ser. ICSR, 2002, pp. 62–77.

[9] K. C. Kang, S. G. Cohen, J. A. Hess, W. E. Novak, and A. S. Peterson, "Feature-oriented domain analysis (foda) feasibility study," Software Engineering Institute, Carnegie-Mellon University, Pittsburgh, PA, USA, Tech. Rep., 1990.

[10] W. K. Kelm, A. V. d. Mello, and M. Bernardino, "The state of practice in requirements engineering in the development of mobile applications," in *17th Brazilian Symposium on Software Quality*, ser. SBQS, 2018, pp. 210–219.

[11] D. Zowghi and C. Coulin, "Requirements elicitation: A survey of techniques, approaches, and tools," in *Engineering and Managing Software Requirements*, C. Wohlin *et al.*, Eds. Springer, 2005, pp. 19–46.

[12] K. J. Sternberg, M. E. Lamb, P. W. Esplin, and L. P. Baradaran, "Using a scripted protocol in investigative interviews: A pilot study," *Applied Developmental Science*, vol. 3, no. 2, pp. 70–76, 1999.

[13] A. F. Osborn, *Applied imagination: Principles and procedures of creative thinking*. Scribner's, 1953.

[14] P. Grünbacher and N. Seyff, "Requirements negotiation," in *Engineering and Managing Software Requirements*, C. Wohlin *et al.*, Eds. Springer, 2005, pp. 143–172.

[15] L. Macaulay and J. Mylopoulos, "Requirements engineering: an educational dilemma," *Automated Software Engineering*, vol. 2, no. 4, pp. 343–351, 1995.

[16] M. Daun, A. Salmon, B. Tenbergen, T. Weyer, and K. Pohl, "Industrial case studies in graduate requirements engineering courses: The impact on student motivation," in *27th IEEE Conference on Software Engineering Education and Training*, ser. CSEE&T, 2014, pp. 3–12.

Authorized licensed use limited to: IEEE Xplore. Downloaded on November 01,2020 at 18:08:15 UTC from IEEE Xplore. Restrictions apply.